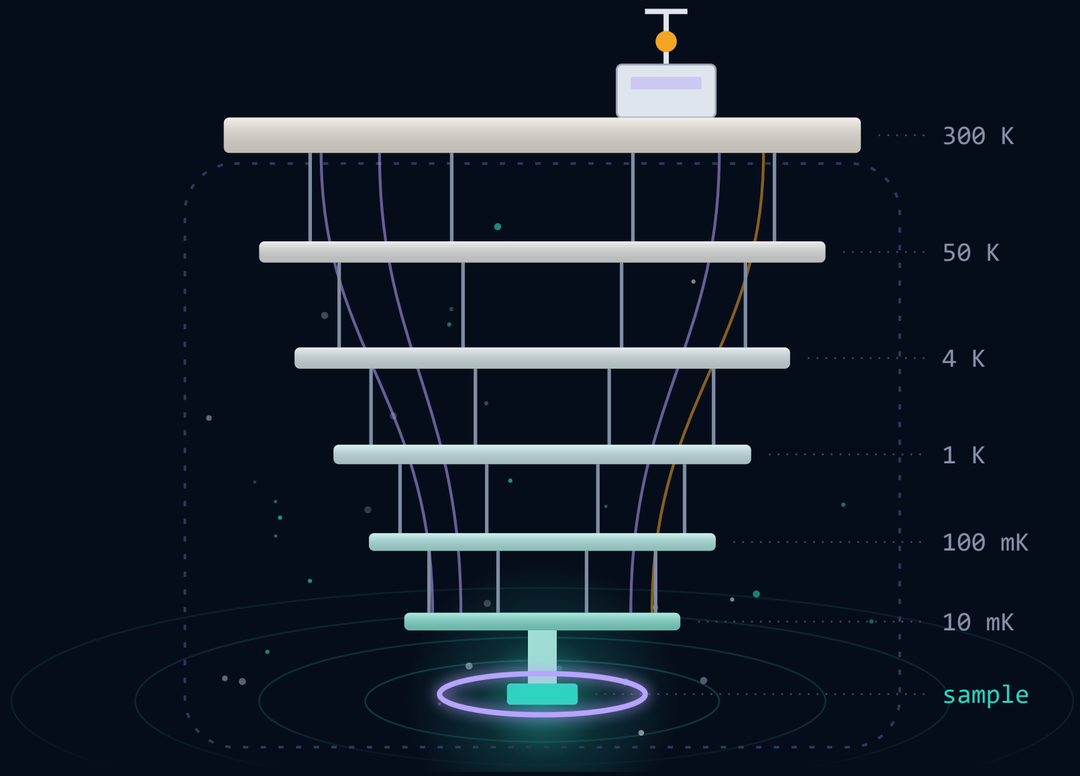
VIBECHECK · THE USABLE FORMAL METHODS
HACKATHON · SAN FRANCISCO · NOV 2026

CryoLean

Cryo, made simple.

20 industries **220 machine setups** **1 Lean checker**



Hanah Rahman · Nicholas Brandenburg

hrahman12.github.io/CryoLean · github.com/hrahman12/CryoLean · generic cryostat drawn in code, no real model

/ SCALE · ONE CHECKER, EVERY LAB

Twenty industries. One unchanged proof.



44,880

experiment-machine pairs decided (204 × 220)

3,335

sourced records, each with URL, location and date

NEW INDUSTRY = NEW DATA

Adding one left the Lean core unchanged. A theorem proves the label can't change a verdict.

`verdict_industry_irrelevant`

site/matrix_data.json (180 pairings, best-match machines) · site/data.json (44,880 pairs) · data/evidence (3,335 records) · docs/examples/demo_c_new_industry_check.json (core hash before = after) · lean/CryoLean/Soundness.lean:1024 · coverage is only what has been researched

/ WHAT WE ASK

Help us turn unknowns into answers.



The website's Demos tab · every verdict from the Lean checker

WE ASK

Makers: sourced specs. **665 of 2,832** entries are still UNKNOWN. **Labs:** real experiment needs.

NEXT WE PROVE

The **5** coupled physics models still UNSUPPORTED (8 of 13 validated).

TRY IT NOW

```
hrahman12.github.io/CryoLean
github.com/hrahman12/CryoLean
python -m cryolean demo
```

Lean checks plans against sourced specs; it does not certify machines · data/ (2,832 capability entries) · data/physics/coupled_models.json · 3D models illustrative